

Question	Answer	Marks
11(a)	discrete amount/quantum/packet of <u>energy</u>	M1
	of electromagnetic radiation	A1
11(b)(i)	energy = hc/λ	C1
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8)/(0.51 \times 10^6 \times 1.60 \times 10^{-19})$ $= 2.4 \times 10^{-12} \text{ m}$	A1
11(b)(ii)	$p = h/\lambda$ $= (6.63 \times 10^{-34})/(2.44 \times 10^{-12})$ or $p = E/c$ $= (0.51 \times 1.60 \times 10^{-13})/(3.00 \times 10^8)$	C1
	$p = 2.7 \times 10^{-22} \text{ N s}$	A1
11(c)(i)	$E = c^2 \Delta m$	C1
	$\Delta m = (0.51 \times 1.60 \times 10^{-13})/(3.00 \times 10^8)^2$ $= 9.1 \times 10^{-31} \text{ kg}$	A1
11(c)(ii)	(momentum is conserved so) nucleus must have momentum in opposite direction to photon	B1

9702/41

Cambridge International AS/A Level – Mark Scheme

October/November 2018

PUBLISHED

Question	Answer	Marks
12(c)	unstable nucleus	D1